

1 Project Description

The final project should be a study of some area of RL of interest to you. For example, you might develop a game playing RL agent for version of poker, go or some other game. The game playing agent might be based on any of the methods we've discussed in class, will discuss in class or anything else that is RL. Alternatively, you might use a common video game (e.g., Atari, Doom or Starcraft) or robotics domains that have or have not been studied before in RL. It doesn't matter which algorithm or application you choose to study. The key requirement is that you think about how the methods we studied in class, and other methods that we did not study, can be applied. Some examples include:

- Recreate algorithm and results from a paper (analyzing whether it produced the same results as expected)
- Compare and analyze some algorithms we studied on a problem of your choice
- Implement an algorithm, run and analyze on a new domain (along with some baseline algorithm from the field)
- Improve an algorithm and compare with original version

You just want to make sure that the algorithms and domain are sufficiently sophisticated (e.g., you don't want to just do Q-learning on a grid-world). Specifically, you shouldn't use any algorithm that are already implementing in some homework. Conversely, some projects are ambitious or time-consuming (e.g., deep reinforcement learning and LLM-based methods often require a long time to train and debug), so make sure you schedule the work to make sure you have something to show at the end.

We encourage students to work in groups of two (or potentially three), but you may also work alone. Projects with three or more students need explicit permission before a project proposal can be submitted. In any case where there is more than one student, each student should work on a different sophisticated algorithm (that aren't simple variants of each other).

2 Timeline and Deliverables

2/23/2026 Project proposal due. Please submit a one or two-page document (as a pdf) describing a proposed problem and solution via Canvas. We will review all project proposals. Some projects will be given the go-ahead via email while we will ask to meet with other project groups. The proposal must show that you have read background material on your topic and are qualified to undertake what you propose to do. It should include full references for the papers and other sources that you have consulted and that will form the foundation for your work. If you are working in a group, each student should work on a different algorithm and what each student

1 项目说明

最终项目应当是你感兴趣的强化学习 (RL) 领域的研究课题。例如, 你可以开发一款基于强化学习的扑克、围棋或其他游戏的智能体。这个智能体可以基于我们在课堂上讨论过的方法、即将讲解的内容, 或是其他任何与强化学习相关的技术。当然, 你也可以选择常见的电子游戏 (如雅达利、毁灭战士或星际争霸) 或机器人领域——无论这些领域是否在强化学习中被研究过。无论选择哪种算法或应用场景, 关键在于思考如何将课堂所学方法以及未涉及的其他技术进行实际应用。具体示例包括:

- 根据论文中的算法及结果进行复现 (分析其是否产生与预期相同的结果)
- 比较并分析我们针对您所选问题研究的若干算法
- 在新领域中实现、运行并分析算法 (同时使用该领域的某些基准算法)
- 改进算法并与原版进行比较

关键是要确保算法和领域足够复杂 (比如不能简单地在网格世界上做Q-learning)。具体来说, 千万别用那些已经在作业里用过的现成算法。另外, 有些项目既耗时又复杂 (比如深度强化学习和基于大语言模型的方法, 往往需要长时间训练调试), 所以要合理安排进度, 确保最后能拿出成果展示。

我们鼓励学生两人 (或最多三人) 组成小组合作, 但也可选择独立完成。若项目涉及三人及以上成员, 则需事先获得明确许可方可提交提案。在多人协作的情况下, 每位成员应分别开发不同的复杂算法 (且不可为彼此的简单变体)。

2 时间表和可交付成果

2026年2月23日项目提案截止日期。 请通过Canvas平台提交1-2页的PDF文档, 详细阐述研究课题及解决方案。所有提案将统一审核, 部分项目将通过邮件快速通过, 其余需安排项目组面谈。提案需证明你已充分研读相关背景资料, 并具备开展研究的资质。请完整列出所有参考文献, 这些资料将作为研究基础。若为团队合作, 请每位成员分别负责不同算法的开发工作。

will work on should be explicitly listed in the proposal. It must specify (using the same titles and questions as below):

1. Problem description: What problem are you solving? Describe the problem *formally* from a computational perspective. What are the states, actions and rewards? What simulator or domain are you using (exactly)? Why is it interesting? Does it already have an interface for RL? Have other researchers already used it?
2. Algorithms: What algorithms do you use (exactly)? What baseline methods will you compare to? Why are these algorithms appropriate? How are these algorithms typically used, and how are you using them? Have other people use similar algorithms to solve your problem before?
3. Results: What results do you expect to show? What comparisons will you do? Are there risks for not getting all the results? If so, what will you do about it?

4/22/2026 Final project due. The papers should be written using the AAAI format (for the AAAI Conference on Artificial Intelligence): <https://aaai.org/authorkit26-1/>. Of course, we don't expect these projects to be submitted to the conference (although you are certainly welcome to!), it is helpful to look at papers from previous years to get an idea how they are written. AAAI papers (and papers from several other conferences) for many years can be found at this link: <http://www.aaai.org/Library/conferences-library.php>. Some sample project reports will also be made available. Also submit the code used to produce your results as a separate link or zip. Your report can be organized differently, but the general organization is the following:

1. Abstract: A short summary of what problem you are solving, how you solved it and what the results are.
2. Introduction: A longer description motivating the problem and solution method.
3. Background: Any background information needed to understand the methods used in the project (e.g., a description of a general search problem or some simpler algorithms that you build off of).
4. Related work: What other methods could be applied to your problem, why didn't you use them and how they relate to your method.
5. Project description: What you actually did in formal detail (with algorithms, equations, etc.).
6. Experiments: A description of how you chose these experiments, how the experiments were run, what the results were and why you got these results (Under what circumstances does the algorithm solve your problem successfully? When does it fail?) Again, they should be formal, often with graphs. These results could also include analysis such as a comparison of different methods and performance on different variants of the problem.
7. Conclusion: A summary of the results and what you learned by trying to complete this project.

Project reports must be detailed and self-contained, explaining the problem, methods and results.

Note: there can not be any extensions on the final project!

应在提案中明确列出工作内容。必须使用与下文相同的标题和问题进行具体说明：

1. 问题描述：您要解决的问题是什么？请从计算角度正式描述该问题。问题包含哪些状态、动作和奖励？您使用的是哪种模拟器或领域（具体说明）？该问题为何具有研究价值？是否已具备强化学习接口？是否有其他研究者使用过该问题？
2. 算法：您具体采用哪些算法？将与哪些基准方法进行对比？这些算法为何适用？这些算法通常如何应用，您又是如何使用的？此前是否已有其他研究者采用类似算法解决过您的问题？
3. 结果：您预期获得哪些结果？将进行哪些比较分析？是否存在无法获得全部结果的风险？若存在此类风险，您将采取何种应对措施？

2026年4月22日为最终项目截止日期。论文需按照AAAI格式撰写（适用于AAAI人工智能会议）：<https://aaai.org/authorkit26-1/>。虽然我们不强求提交这些项目（当然欢迎投稿！），但参考往届论文的写作方式会很有帮助。多年来AAAI会议论文（及其他多个会议论文）可在此链接查阅：<http://www.aaai.org/Library/conferences-library.php>。部分项目报告样本也将同步开放。同时请将生成结果的代码作为独立链接或压缩包提交。报告结构可灵活调整，但基本框架如下：

1. 摘要：简要概述您所解决的问题、解决方法及结果。
2. 引言：对问题及解决方案方法进行更详细的描述。
3. 背景：理解项目所用方法所需的任何背景信息（例如，对一般搜索问题的描述或基于其构建的某些简化算法）。
4. 相关研究：针对您的问题，还可采用哪些其他方法？为何未选用这些方法？它们与您所采用的方法有何关联？
5. 项目描述：您实际执行的正式详细工作（含算法、方程等）。
6. 实验部分：需详细说明实验方案的选择依据、具体实施过程、实验结果及其成因（算法在何种条件下能成功解决问题？何时会出现失效？）。所有内容均需采用规范化的表述方式，建议辅以图表辅助说明。此外，可包含不同方法的对比分析，以及算法在问题不同变体上的性能表现评估。
7. 结论：通过尝试完成本项目，对结果及所获知识进行总结。

项目报告必须详尽且自成一体，需说明问题、方法及结果。**注：**最终项目不得有任何扩展！